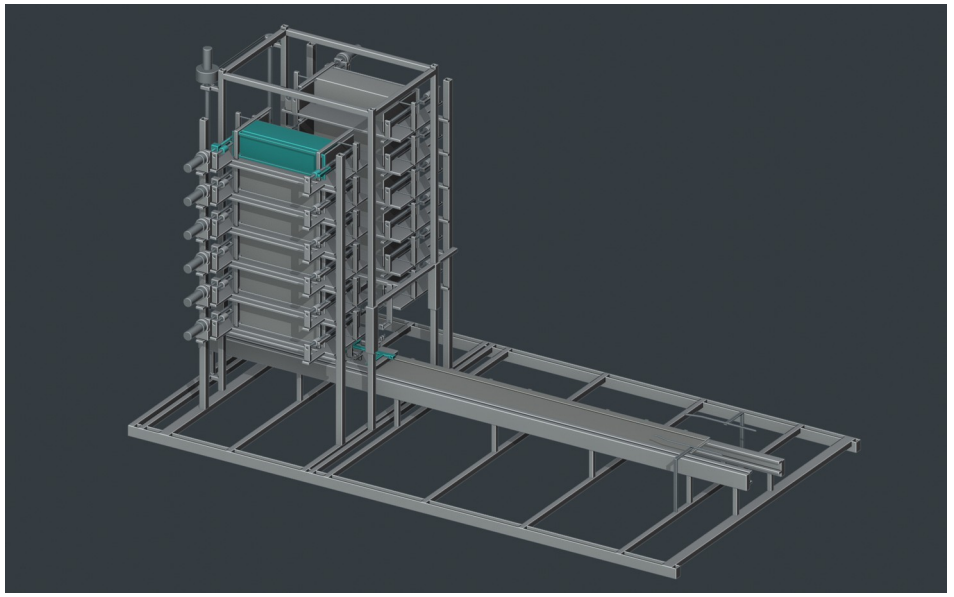
DEPARTMENT OF MECHANICAL ENGINEERING

VOLLEY

Design and Computational Evaluation of a Commanded Electromagnetic CubeSat Deployment Mechanism

GEN5 ACR1 | COMPUTATIONAL DESIGN STUDY



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Presentation outline

01 Motivation and prior work Introduction, literature, research gap and objectives

02 Method and progress Controlled configuration and dependent numerical models

03 Mechanism and results Payload custody, finite motor, source, structure and mission

04 Conclusions and next steps Study outcome, Gen6 prototype and references

Commanded release adds a local delivery variable

DEPLOYMENT EVENT - SCHEMATIC



Concept only; arrows are not to scale.

- Host manoeuvres establish the carrier state.
- A release command changes the payload initial condition relative to the host.
- Retention, guidance, arrest and host recoil complete the system boundary.

Research question

Can a twelve-payload system complete a coupled, reusable deployment cycle?

Sources: JAXA deployment ICD; Cal Poly CDS Rev. 14.1; Foster et al. (2018); Vallado (2013)

Prior work establishes the mechanisms and comparison context

Published work	Evidence / interface	Implication for VOLLEY
Zhao et al., 2022 Stacked EM deployment	Gravity-compensated tests; stacked CubeSat release	Distinguish published measurements from the present computational results.
Zhao et al., 2025 High-volume storage	Receiving guides and magnetic transfer; payload grooves and iron patches	Define the passive payload interface and receiving custody explicitly.
Feng et al., 2025 EM reachable domain	Higher-speed launcher and mission-domain analysis	Couple the selected operating regime to its energy and mission assumptions.
Halbach; finite-motor studies	Permanent-magnet field shaping and longitudinal end effects	Use finite magnet and conductor geometry at launch and arrest.

Study focus: integration and same-revision evaluation of a retained-carrier architecture.

Sources: Zhao et al. (2022, 2025); Feng et al. (2025); Halbach (1980); Yamaguchi & Morishita (2023)

Research gap: closing the complete retained-carrier cycle

Who holds the payload?

A complete custody path through cell opening, receiving, docking and commanded departure.

Can the source deliver it?

Finite force, actual moving mass, current dynamics and post-release arrest evaluated together.

What does the host retain?

The reusable carrier, installed hardware, power allowance and recoil carried into mission comparison.

One design links payload custody, electromechanics and mission resources.

Positive custody precedes each handoff; the same carrier serves launch, arrest and return.

Objectives and controlled reference requirements

12 × 4 kg
Identical 3U envelope surrogates

1–3 m/s
Analysed commanded-speed cases

2250 mm
Guide-rail allocation

210 mm
Post-release arrest allocation

- Define continuous payload custody and a reusable carrier cycle.
- Evaluate finite motor force with the represented winding and magnet geometry.
- Couple source, release and arrest; assess structural and mission consequences.

Sources: frozen CAD manifest; PARAMETERS.json; coupled operating-study definition

One source chain connects the machine to the mission

1

Define the machine

Geometry, mass and custody

2

Solve the shot

Finite force + source + arrest

3

Check the structure

Loads, gap and declared budgets

4

Compare the mission

Same targets; full retained mass

Cross-check each model at its own boundary.

Native / STEP geometry • field and current formulations • refined shot / native orbit replay

Sources: accepted ACR1 CAD, force/current, source-coupled shot, structural and mission records

Timeline and work progress

01

Problem and literature

Complete

02

Gen5 ACR1 definition

Complete

03

Coupled computational study

Complete

04

Report and presentation

Academic edition ED3

05

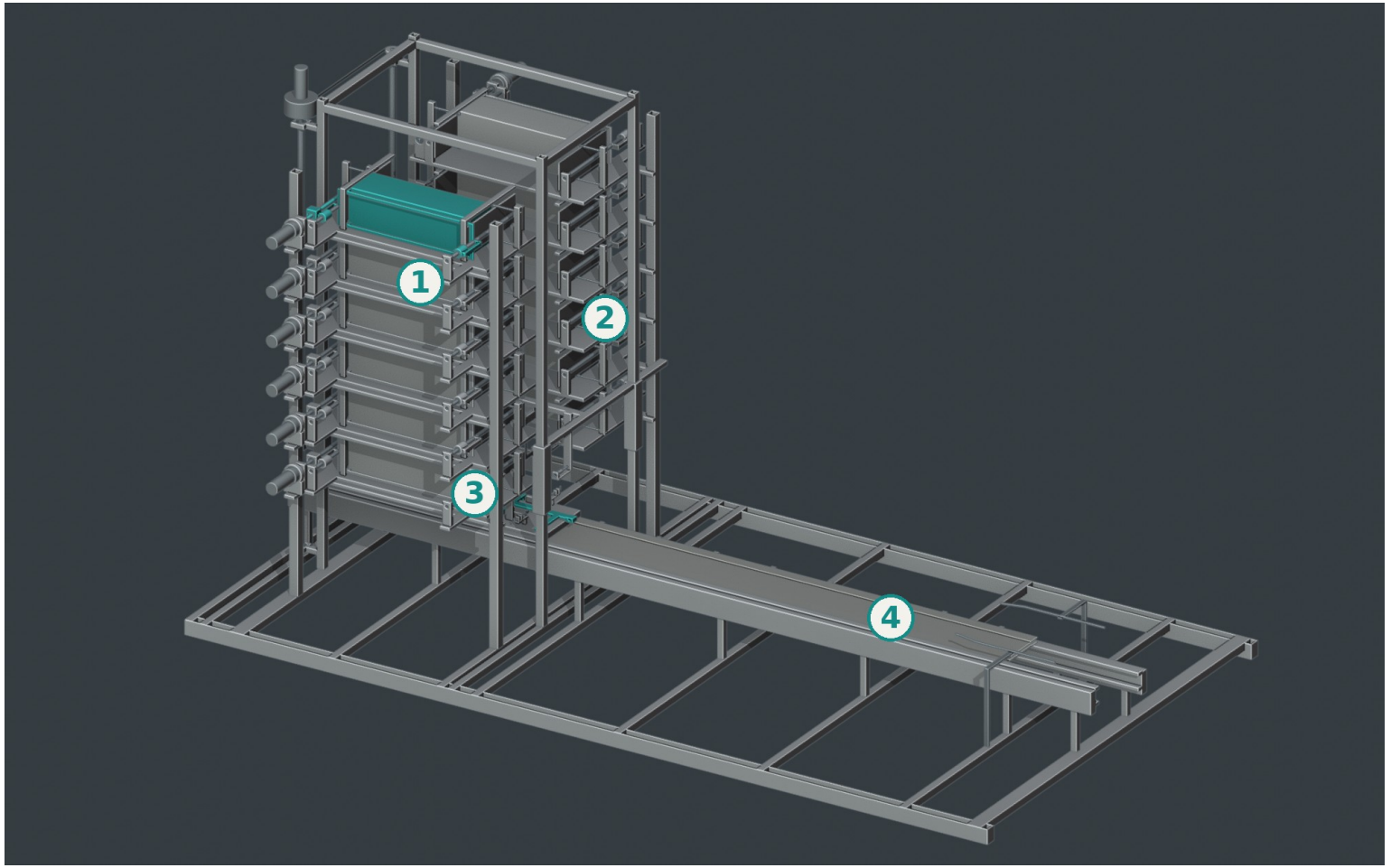
Gen6 instrumented prototype

Future work

Completed outcome

A controlled computational design reference with defined operating cases and a hardware validation programme.

Reference architecture: fixed banks and one captive pallet



1 Storage · 2 Lift · 3 Carrier/pallet · 4 Stator + track

2 × 6 cells
Fixed, individually retained storage

13.128 kg
Retained moving carrier + pallet

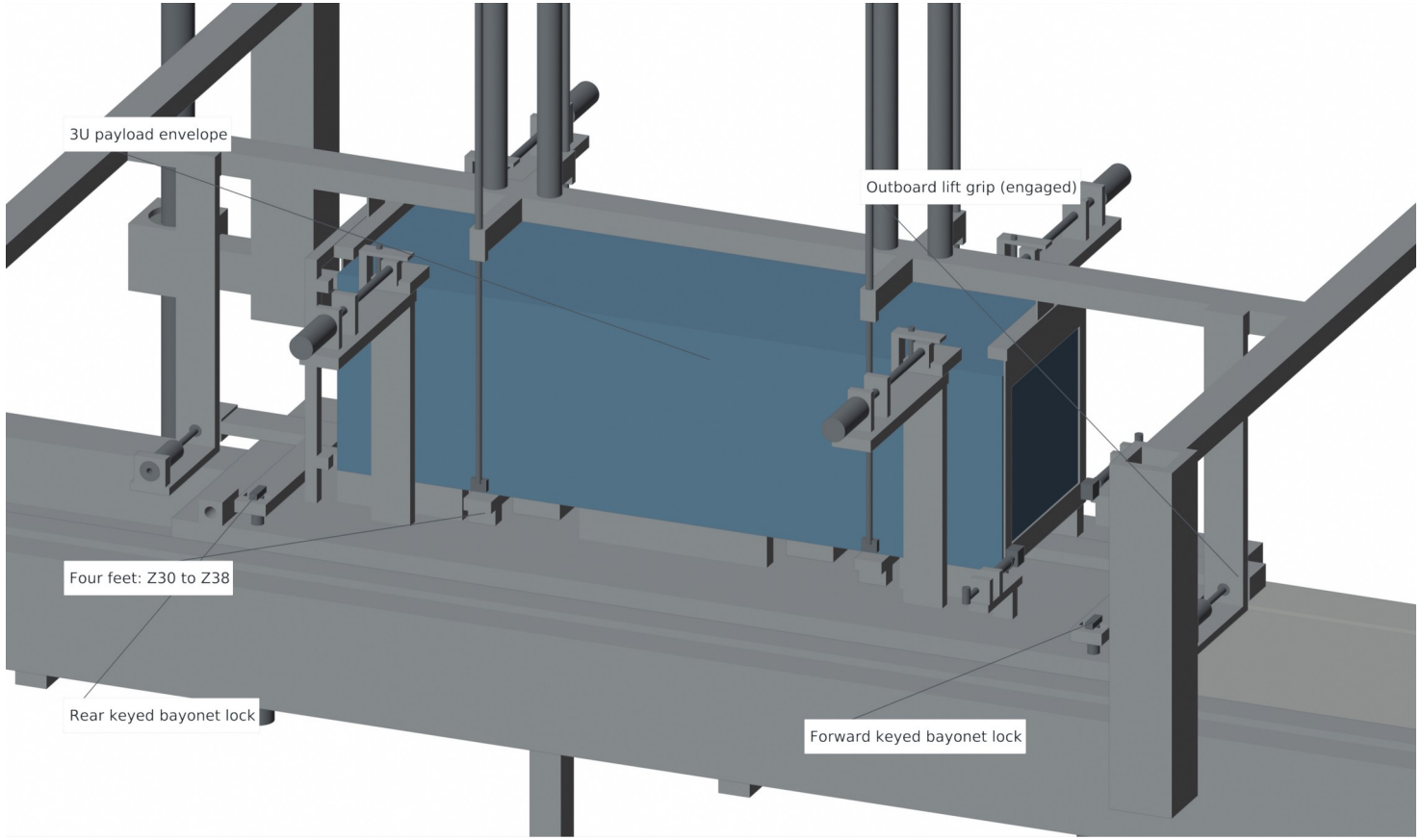
149.557 kg
Represented dry CAD mass

+7 kg power/control allowance

Magnet arrays stay on the carrier.

CAD-derived render: frozen ACR1 home pose; all 387 represented parts; declared reference components

Capture before release: one pallet, two holders



1 Lift holds

The lift grips retain the pallet while it lands.

2 Carrier captures

Caps and both bayonet locks prove receiving custody.

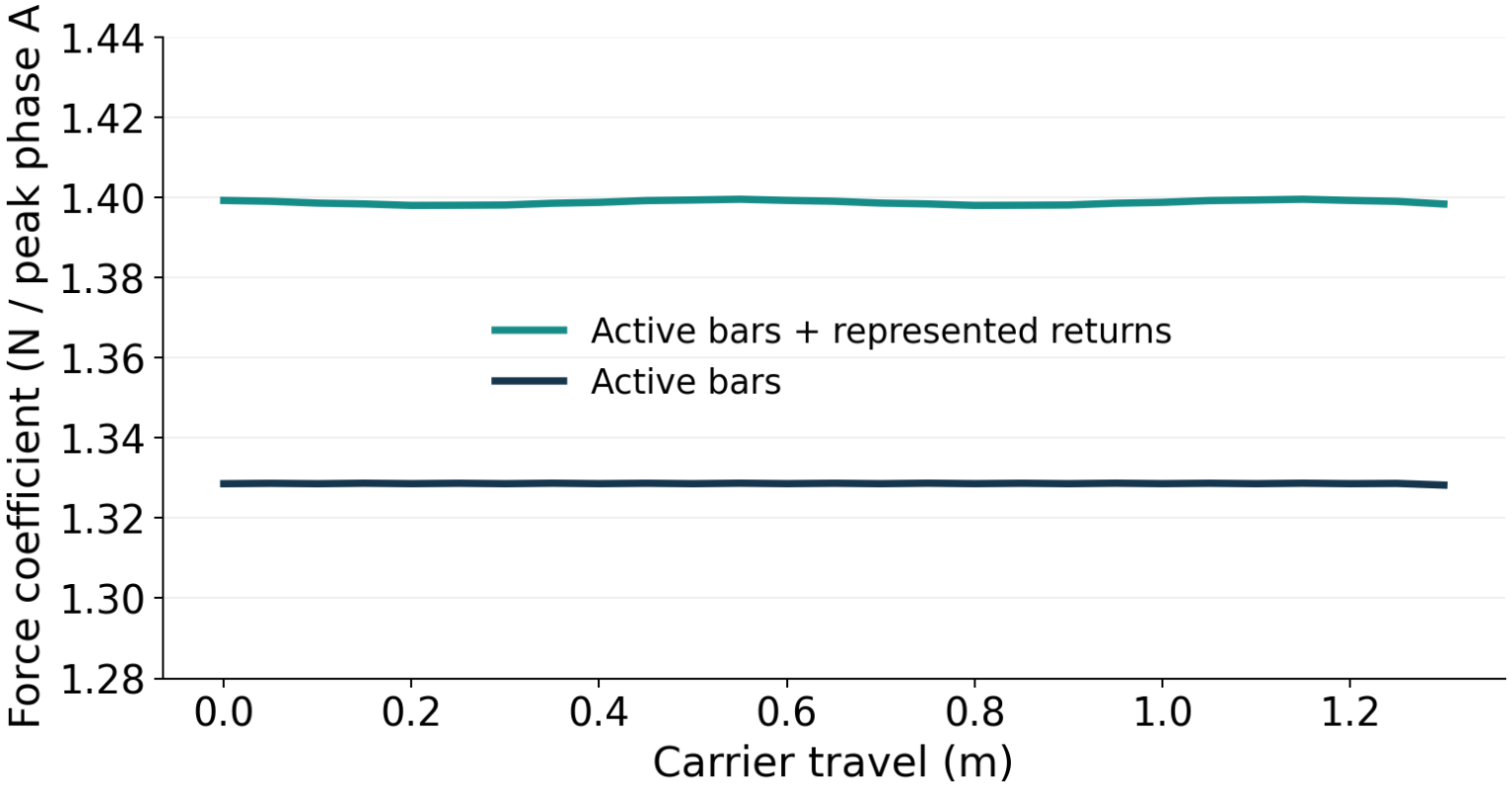
3 Lift clears

Only then do the lift grips withdraw and the lift rise.

The pallet stays on the carrier through launch, arrest and return.

CAD-derived ACR1 handoff pose; banks, host ladder and fixed lift frame hidden to expose the interfaces

Finite geometry gives the force used by the shot model



1.399 N/A

Interior model, including represented returns

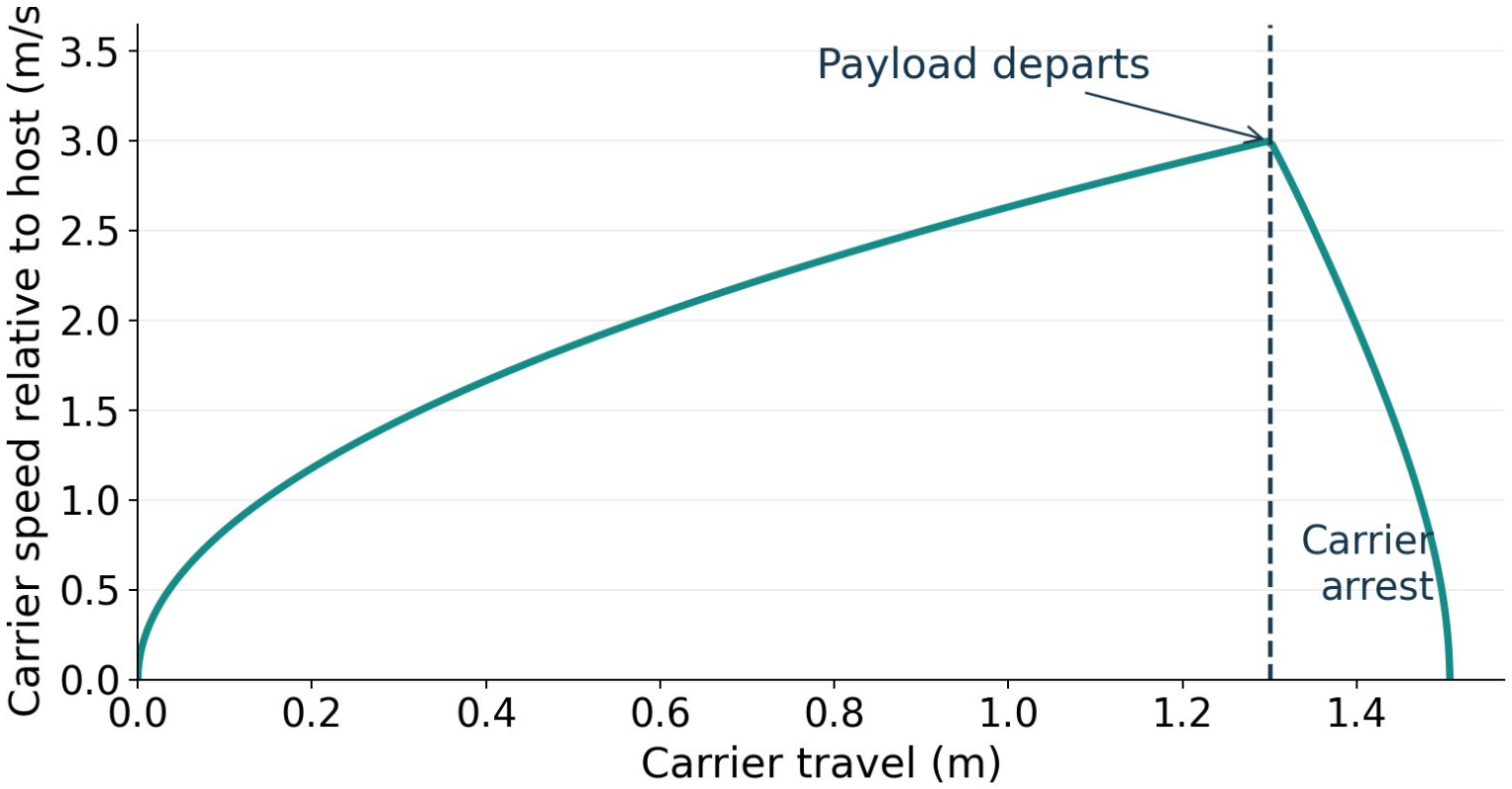
0.172%

Local actual-current FEM difference under the same permanent field

42 + 42 discrete magnets • 216 active bars • finite conductor and return geometry

Source: FINITE_MOTOR.json; CURRENT_FIELD_FORCE.json | coefficients use peak phase current

The carrier accelerates, releases, then arrests



3.000 m/s

Modeled speed at departure

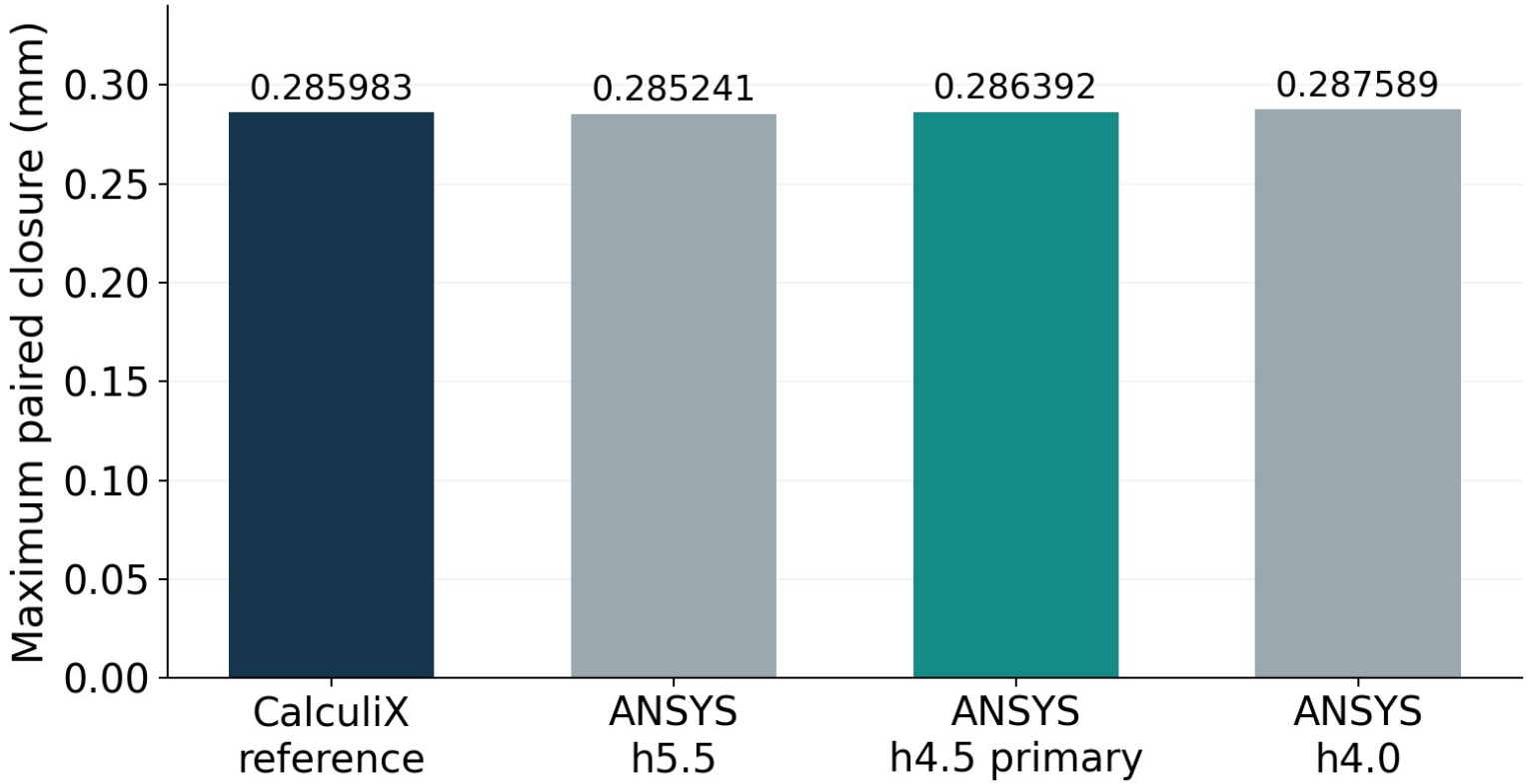
379.7 J

Nominal source draw

6 / 6 principal cases pass the 5 µs timestep audit.

Release near 1.300 m → carrier stops at 1.507 m within the 210 mm arrest allocation.

A second solver agrees under the matched chassis boundary



+0.143%

Primary ANSYS difference from CalculiX

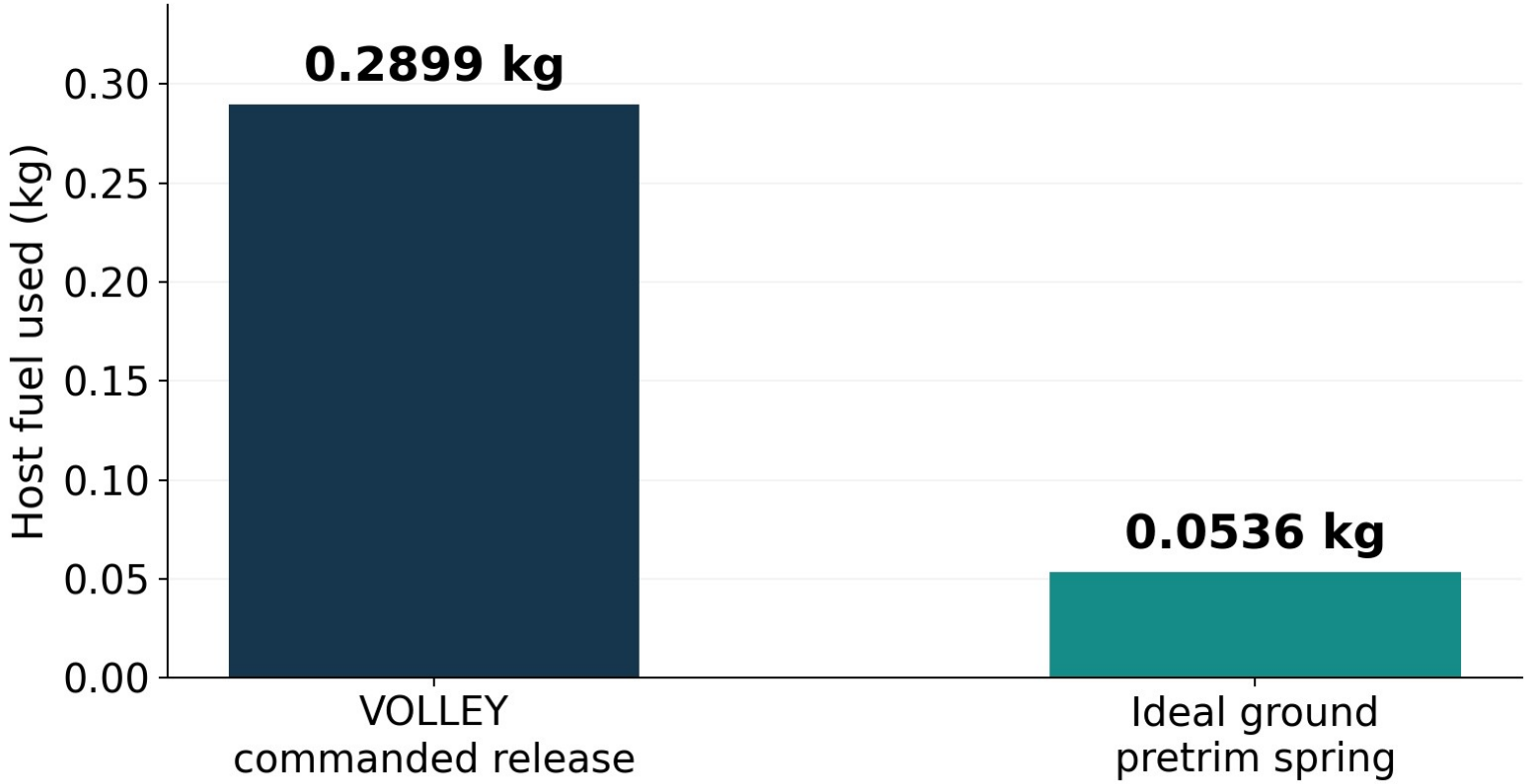
Three ANSYS meshes

Sensitivity evidence; no formal error bound.

Opposed 3.010 kN patches • six scalar numerical restraints • native-result sample readback

Sources: CalculiX reference and corrected ANSYS native-result audit | same bare-chassis loads and restraints

The preplanned mission trade favours ideal ground pretrim



84.557 kg

Additional retained device mass in the study

156.557 kg VOLLEY
72 kg spring class

Spring mass is a declared class-study assumption.

Same 500 kg wet bus + 48 kg payload; device mass and release/burn policies differ.

Conclusions of the computational design study

A complete reference cycle is defined

Fixed cells, guided feed, positive pallet capture, commanded release, active arrest and return.

The declared low-speed cases are computationally supported

1–3 m/s cases analysed; the principal 3 m/s matrix passes a reviewed shared-solver timestep audit.

The retained system burden is visible

149.557 kg represented dry CAD plus 7 kg allowance; no resource advantage over ideal pretrim in the stated mission comparison.

Gen5 ACR1 completes the computational reference for physical validation.

Gen6: an instrumented mixed 1U / 3U physical prototype

1 Characterise

Force / back emf

Resistance and inductance

Source ESR and switching

2 Correlate custody

Pusher and guide forces

Pallet and lock repeatability

Payload contact and angular rate

3 Demonstrate the cycle

Release and active arrest

Repeated thermal duty

Mixed-envelope handling

Further scaling requires fresh stroke, acceleration, energy, arrest and heat-rejection budgets.

Selected references

01 Cal Poly, CubeSat Design Specification Rev. 14.1; JAXA, JEM small-satellite deployment ICD.

02 Zhao et al. (2022), stacked CubeSat deployer. Remote Sensing 14, 4205. doi:10.3390/rs14174205.

03 Zhao et al. (2025), high-volume storage transport. Aerospace 12, 466. doi:10.3390/aerospace12060466.

04 Feng, Yang and Wu (2025), electromagnetic launcher and reachable domain. IJAE 3000765. doi:10.1155/ijae/3000765.

05 Halbach (1980), permanent-field design; Yamaguchi and Morishita (2023), finite-motor end effects. Full identities in report.

06 Computational methods: FreeCAD, Magpylib, Gmsh, CalculiX, Orekit, Blender; corrected ANSYS static source supplement (2026).

Questions